Dielectric Insertion: Fields, Voltage, and Causality

Your cancellation argument is correct: for ideal infinite sheets at fixed free charge, inserting the dielectric does not change the electric field in the air gaps. However, **that field is not the entire electromagnetic field of the capacitor–battery system**. The apparent contradiction arises when an infinite-sheet *electrostatic* result is used to rule out the fields needed for a *time-dependent circuit response*.

Saying "the battery keeps the voltage constant" gives the final equilibrium condition, but does not explain how the battery responds causally.

1 A voltage change does not require a field change at the plates

Let σ be the free surface-charge density on the positive plate, d the plate separation, b the dielectric thickness, and κ its relative permittivity. In the ideal planar geometry, at fixed charge,

$$E_{\text{air}} = \frac{\sigma}{\varepsilon_0}, \qquad E_{\text{dielectric}} = \frac{\sigma}{\kappa\varepsilon_0}.$$

The dielectric's two infinite, uniformly charged bound surfaces cancel each other's fields outside the slab, exactly as you argued. Nevertheless, the positive plate's potential relative to the negative plate is

$$\begin{aligned} V &= \int_{\text{positive plate}}^{\text{negative plate}} \mathbf{E} \cdot d\boldsymbol{\ell} \\ &= E_{\text{air}}(d-b) + E_{\text{dielectric}} b \\ &= \frac{\sigma}{\varepsilon_0}\left(d - b + \frac{b}{\kappa}\right). \end{aligned}$$

This is smaller than the original $\sigma d/\varepsilon_0$ because the field is smaller *along part of the path between the plates*.

There is no requirement for charges on the plates to experience a different local force just because that integral changes. A potential difference is not an additional local force that the charges must be "notified" about.

For the disconnected capacitor in the textbook, that is the whole story: the voltage decreases, the free charge stays fixed, and no battery needs to respond. A connected battery requires a more complete description.

2 The battery responds to fields along its connection

In an actual capacitor–battery circuit, inserting and polarizing the dielectric involves moving bound charges and polarization currents. These, together with the responding charges on the conductors, produce a changing electromagnetic field.

The disturbance propagates through the apparatus at a finite speed. It changes local electric fields along the leads and at the battery terminals, allowing current to flow.

The field driving current along a wire is its component along the wire. Your cancellation calculation concerns the field normal to the plates in the ideal planar region. It does not establish that the field along the battery connection remains unchanged.

Moreover, cancellation in an initial or final electrostatic state does not establish cancellation throughout the transient. During that transient,

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi_B}{dt},$$

where Φ_B is the magnetic flux through the loop. The purely electrostatic picture is therefore not the complete description of the response.

The battery does not remotely measure the dielectric and decide to compensate. Its charges respond locally to electromagnetic and electrochemical forces.

3 What changes if the plates are exactly infinite?

Take the idealization completely seriously:

- The plates and dielectric are exactly infinite in the directions parallel to the plates.
- Everything is translationally symmetric in those directions.
- The free surface-charge density is fixed.

Then the dielectric can indeed change the potential difference without changing the external electrostatic field. But **that mathematical model contains no localized battery connection whose response it can describe**.

A finite wire or contact, a hole through a plate, or a battery placed between the plates breaks the symmetry. Alternatively, a connection going *around* truly infinite plates has no finite length. One cannot assume exact planar symmetry while treating the geometry of a localized connection as irrelevant.

A closed-loop argument makes the missing information explicit. Compare the *fixed-charge electrostatic states before and after insertion* for a *finite* capacitor. Draw a closed path through the capacitor gap and back around the outside. Write $\Delta\mathbf{E}$ for the difference between the two static fields. Since both states are electrostatic,

$$0 = \oint \Delta\mathbf{E} \cdot d\boldsymbol{\ell} = \underbrace{\int_{\text{gap}} \Delta\mathbf{E} \cdot d\boldsymbol{\ell}}_{\text{nonzero}} + \int_{\text{outside}} \Delta\mathbf{E} \cdot d\boldsymbol{\ell}.$$

Consequently, **the field cannot remain unchanged everywhere along the outside return path**. Nonuniform fields near and beyond the edges provide the compensating contribution, even when the bulk air-gap field is essentially unchanged.

For infinite plates, there is no finite outside return path of that kind. The apparent puzzle comes from removing the edges and connection geometry while still expecting the resulting model to explain communication with an external battery.

The infinite-sheet approximation accurately describes the bulk electrostatics; it is not a complete model of the circuit's causal response.

4 Once the battery supplies charge, the air-gap field changes too

The statement "the air-gap field is unchanged" is conditional on **fixed** σ. For a battery maintaining the original voltage V_0, the final equilibrium in the parallel-plate approximation instead satisfies

$$V_0 = \frac{\sigma_{\text{final}}}{\varepsilon_0}\left(d - b + \frac{b}{\kappa}\right).$$

Thus,

$$\sigma_{\text{final}} = \frac{\varepsilon_0 V_0}{d - b + b/\kappa}, \qquad E_{\text{air,final}} = \frac{V_0}{d - b + b/\kappa}.$$

For $b > 0$ and $\kappa > 1$, this air-gap field is larger than the original V_0/d. These equations describe the final state; the electromagnetic transient is how a real circuit approaches it.

Bottom line

You are right that causal influence must be electromagnetic. What fails is the inference that an unchanged bulk air-gap field at fixed charge means no field change can reach the battery.

An unchanged field in one region does not imply unchanged fields throughout the circuit, especially during insertion. The ideal infinite-sheet calculation omits the connection geometry and transient fields needed to answer the communication question.